

EDUCATION

University of Toronto

PhD, Computer Science

Incoming Fall 2025

Research Area: Computational Imaging, Computer Vision, and AI for Science

MEng, Mechanical and Industrial Engineering

2024 - 2025

Coursework: Machine Learning, Deep Learning, Automated Reasoning with Machine Learning, Computational Imaging, Geophysical Fluid Dynamics

Honours BS, Mathematics

2019 - 2024

With High Distinction; **3.85 final 2-year GPA**

Coursework: General Relativity (graduate level), Hamiltonian Mechanics, Human Physiology, Computability Theory, Differential Geometry, Analysis

RESEARCH

Graduate Student Researcher, advised by Hamed Ibrahim

Sep 2024 - Present

University of Toronto Department of Mechanical and Industrial Engineering

- Developed a data analysis library in Python to investigate tropopause dynamics from global reanalysis and radiosonde datasets.
- Designed and implemented custom numerical algorithms for computing atmospheric variables and double tropopause detection.
- Designed and deployed a web application for interactive radiosonde data analysis and visualization (igrapytools.com). Implemented with Python, Flask, JavaScript.

Research Trainee, advised by Morgan O'Neill

May 2024 - Aug 2024

University of Toronto Department of Physics

- Significantly enhanced a previous 2.5-layer moist convective shallow water model for simulating polar dynamics of giant planets, by developing a new parameterization for moist convection, rewriting the codebase for parallelization with Numba, adapting the model to Jupiter-like conditions, adding more physical terms, and conducting long-term simulations on the Niagara supercomputer.
- This resulted in a greater than $50\times$ improvement in resolution and time integration, achieving unique geometric results currently being drafted into a paper.
- Proposed the use of machine learning methods for modeling giant planet polar dynamics. Explored various architectures and models including PINNs.

Undergraduate Student Researcher, advised by Emmy Murphy

Jan 2024 - May 2024

University of Toronto Department of Mathematics

- Authored one of the first works to comprehensively review exotic 7-spheres, seven-dimensional manifolds that are homeomorphic but not diffeomorphic to Euclidean spheres.
- Reviewed the necessary tools from algebraic topology and differential geometry, expanded on the original proofs, and presented an equivalent technique to prove the key homeomorphism by directly computing the homology of the manifold.
- Opens as PDF: pnphan.github.io/seniorthesis.pdf

Research Trainee, advised by Milad Lankarany

May 2022 - Aug 2022

University of Toronto Institute of Biomedical Engineering

- Conducted a systematic study of mathematical neuron models, explored relevant optimization algorithms, and implemented various models in MATLAB and Python.

PROJECTS

Hyperspectral Image Superresolution in the Fourier Domain

- Proposed and developed a novel deep convolutional neural network architecture for hyperspectral image super-resolution. This included spatio-spectral attention mechanisms, a unique training scheme with generative adversarial networks, a frequency-domain loss function, and a dual frequency-spatial domain fusion network, achieving superior performance compared to existing methods.
- Paper link: https://www.cs.toronto.edu/~lindell/teaching/2529/past_projects/2024/report/peter-phan.pdf

IGRAT: a Python library for radiosonde data analysis

- Python library for pre-processing data from the Integrated Global Radiosonde Archive (IGRA). Allows users to visualize temporal and spatial patterns, plot atmospheric profiles, and export processed data sets in more standard formats.
- Submitted for publication in *Geoscience Data Journal*
- Library: <https://github.com/pnphan/igrat.git>
- Web interface (first make sure the NCEI servers are up): <https://igrapytools.com/>

TEACHING

TA Teaching Excellence Award Finalist

2023

University of Toronto

- The first and only undergraduate finalist in award history.
- 14 finalists were selected from 324 faculty-sponsored and student-nominated Teaching Assistants across all campuses.

Head Teaching Assistant / Assistant Coordinator

Sep 2022 - Present

University of Toronto

- Head Teaching Assistant for 6 iterations of mathematics courses on introductory proofs, and single and multivariable calculus.
- Managed team of Teaching Assistants, assigned grading allocations, responded to regrade requests and student concerns, assisted in assessment development.

Teaching Assistant

Jan 2022 - Present

University of Toronto

- Teaching Assistant for 15 iterations of various mathematics courses on numeracy, introductory proofs, discrete math, differential equations, and single and multivariable calculus.
- Led weekly tutorials, graded student assessments, conducted weekly office hours, invigilated in-person assessments, provided in-class lecture support.
- Completed over 2000 contracted hours.

Research Assistant

Dec 2022 - May 2024

University of Toronto Institute for the Study of University Pedagogy

- Conducted literature review on psychological process and object conception in mathematics, developed and administered an evidence-based interview protocol to math TAs, analyzed collected data using qualitative research techniques, presented final research poster at the Jackman Research Colloquium.
- Completed various other internal projects in mathematics education and history.

HONORS AND AWARDS

- University of Toronto Department of Physics Summer Research Award (\$9,500) 2024
- Jackman Scholar Award (\$1,000 + accommodations, ~120 selected from ~1200 applicants) 2024
- UofT TA Teaching Excellence Award Finalist (14 selected from 324 candidates) 2023
- University of Toronto National Biology Scholar (placed top 5%, <200 of 3433 contestants) 2018

SKILLS AND INVOLVEMENT

Skills: Python, MATLAB, Machine Learning (PyTorch, TensorFlow, JAX), Numba, $\text{\LaTeX}$, Data Processing (Pandas, SciPy, netCDF, etc.)

Talks/Presentations:

- Jackman Research Colloquium Poster Presentation 2024
- Exotic Spheres: a high-level introduction (University of Toronto Mississauga Math Club) 2023
- Introduction to Field Extensions (University of Toronto Mississauga Math Club) 2022